AI-Powered Intelligent Home Planning & Optimization System

Shevin Rukshan Rathnasinghe
Sri Lanka Institute of Information Technology
Sri Lanka
shevinrukshan12@gmail.com

Vihanga Dinetha
Sri Lanka Institute of Information Technology
Sri Lanka
vihangamdinetha@gmail.com

Ridme Sunkalpani
Sri Lanka Institute of Information Technology
Sri Lanka
ridmisamarathunga2000@gmail.com

Janani Hansika
Sri Lanka Institute of Information Technology
Sri Lanka
jananihansika02@gmail.com

Samantha Rajapaksha
Sri Lanka Institute of Information Technology
Sri Lanka
samantha.r@sliit.lk

Adya Dissanayaka
Sri Lanka Institute of Information Technology
Sri Lanka
adya.d@sliit.lk

***Abstract*—Modern house planning is complex due to limited land, space constraints, and cultural requirements, often requiring costly expert input. This research proposes an AI-powered home planning system integrating land utilization, automated 2D plan generation, interior and exterior optimization, and Vastu compliance evaluation. It uses YOLOv8 and CNN models for room detection, alongside rule- and constraint-based optimization for layouts and furniture placement. Operating on structured JSON data, the system ensures modular integration and efficient processing. Experiments show it provides a cost- and timeefficient solution, offering intelligent decision support without expert intervention.**

***Keywords*—Artificial Intelligence, Smart Home Design, Floor Plan Generation, Space Optimization, Vastu Compliance.**

I. Introduction

Residential planning faces challenges from urbanization, irregular land, and regulations, especially in developing countries, increasing costs and planning time. Traditional workflows, including land analysis, floor plan design, interior/exterior layouts, and cultural compliance, are fragmented, causing poor space utilization and repeated revisions. While AI (YOLO, CNN) enables room detection [3], [4] and rule-/constraint-based methods optimize furniture [5], [6], exterior planning and Vastu compliance remain manual [7]. We propose a unified system with four modules (land utilization and 2D plan generation, AI-based Vastu inspection, interior space optimization, and exterior planning) combining deep learning, rule-based, and constraint-based methods on structured data for efficient, cost-effective planning.

II. Related Work

A. Automated Land Planning and Floor Plan Generation

Early rule-based approaches [8], [9] lack flexibility for complex land, while GAN methods [10] ignore land boundaries and outdoor spaces.a

B. AI-Based Floor Plan Analysis and Room Detection

CNNs perform semantic segmentation and room classification [3], [4], and YOLO improves room detection [11], but most systems lack integration with optimization frameworks.

C. Interior Space Optimization and Furniture Arrangement

Rule- and constraint-based methods optimize circulation and ergonomics [5], [6], with learning-based approaches improving adaptability [12], though prior work treats interiors in isolation.

D. Exterior Space Planning and Cultural Compliance

Outdoor planning research focuses on zoning and paths [13]. Vastu compliance is largely manual; existing approaches [7], [14] lack scalability and AI integration.

E. Summary of Research Gaps

No fully integrated system combines land-use optimization, floor plan generation, interior/exterior optimization, and cultural compliance, highlighting the need for a unified AI-based residential planning framework.

III. METHODOLOGY

A. Overall System Architecture

The system uses a modular yet integrated architecture combining land analysis, spatial optimization, cultural compliance, and visualization. It includes four modules: land utilization with 2D plan generation, AI-based Vastu detection, interior space optimization, and exterior space optimization. Modules communicate via centralized JSON data flows, sharing user inputs to ensure consistent planning. Image inputs are used for visual analysis (floor plans, room detection), processed with metadata. The workflow generates optimized land use, 2D floor plans, interior and exterior layouts, and a Vastu compliance report.

Figure 1 illustrates the overall system architecture and the centralized JSON-based data flow between system modules.

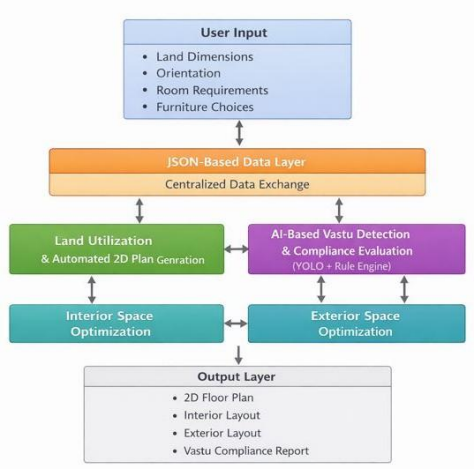


Fig. 1. Overall system architecture showing JSON-based data flow and module interactions

B. Land Utilization and Automated 2D Plan Generation

This module generates feasible 2D layouts from structured land data (dimensions, boundaries, required spaces) using a rule-based model for zoning and spatial feasibility. Requirement-relaxation dynamically adjusts room counts or sizes when constraints exceed land limits. A lightweight ML model predicts layout feasibility. Outputs include optimized land usage and 2D floor plans for downstream modules.

Figure 2 illustrates the workflow of the land utilization and automated 2D plan generation module.

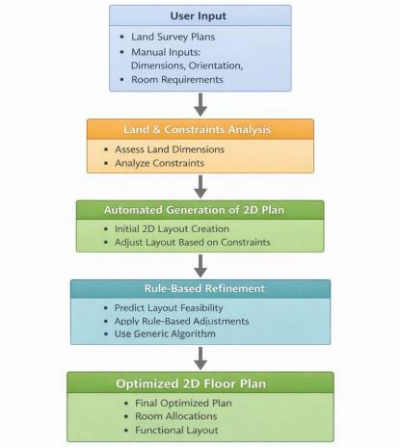


Fig. 2. Workflow of land utilization and automated 2D plan generation module

C. Artificial Intelligence-Based Vastu Detection and Compliance Evaluation

2D floor plan images are normalized and analyzed using YOLO-based room detection, followed by geometric analysis and rule-based Vastu evaluation. The system outputs structured JSON reports with overall status, violations, and recommended actions.

Figure 3 illustrates the processing pipeline of the Vastu detection and compliance evaluation module.

D. Interior Space Optimization Module

The interior space optimization module generates efficient furniture layouts for rectangular and square rooms using a genetic algorithm (GA) that encodes furniture positions, orientations, and spacing as chromosomes. The GA evolves layouts through selection, crossover, and mutation, penalizing constraint violations, while a lightweight ML model provides heuristic guidance based on learned layout patterns.

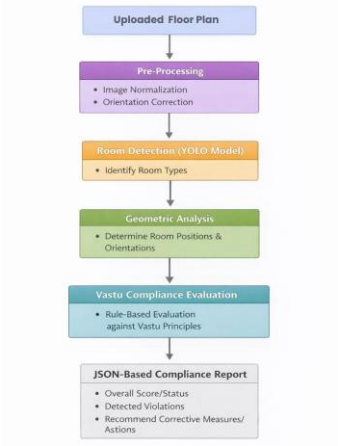


Fig. 3. Processing pipeline of the AI-based Vastu detection and compliance evaluation module

1) Fitness Function Definition: The fitness function enforces hard constraints and rewards desirable spatial arrangements. Constraint violations (furniture outside boundaries or overlapping) return negative penalties: 1000 for boundary violations and 500 times the overlapping area for collisions. Valid layouts receive 10 points per furniture item positioned within 0.1 meters of the nearest wall, encouraging efficient space utilization.

To incorporate learned aesthetic patterns, geometric fitness is combined with an ML-based heuristic score:

$$F_{total}(L) = 0.7 \times F_{geo}(L) + 0.3 \times S_{ML}(L) \qquad (1)$$

where $F_{geo}(L)$ is the geometric fitness score, $S_{ML}(L)$ is a neural network output predicting layout quality from spatial features (room area, aspect ratio, furniture count, centroid distribution), and the weighting factors (0.7, 0.3) prioritize constraint satisfaction while allowing ML-guided aesthetic refinement.

Figure 4 illustrates the workflow of the interior space optimization process, including room type selection, constraint-based validation, fitness evaluation, and iterative layout refinement.

E. Exterior Space Optimization Module

Functional outdoor areas (parking, pathways, landscaping, open spaces) are allocated via rule-based spatial distribution, ensuring coordination with interior layouts and efficient land use.

Figure 5 illustrates the workflow of the exterior space optimization module, including land boundary analysis, functional zone allocation, access rule enforcement, and structured outdoor layout generation.

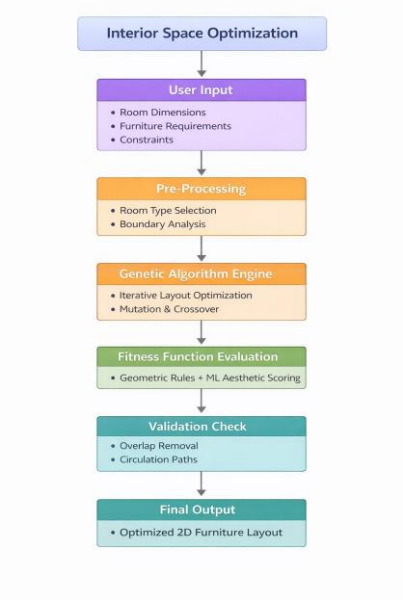


Fig. 4. Workflow of the Interior Space Optimization Module

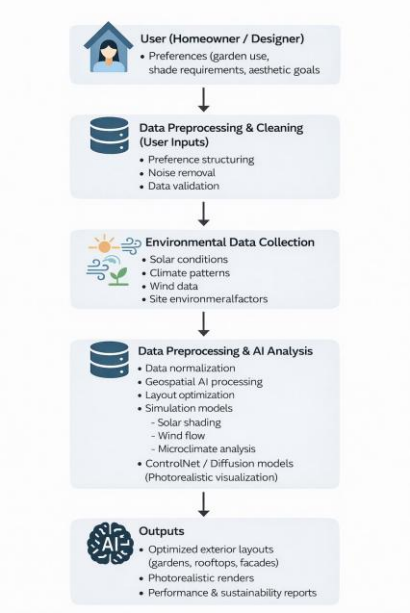


Fig. 5. Workflow of the Exterior Space Optimization Module

F. System Integration and Data Flow

Modules are interconnected via JSON, with outputs serving as inputs for subsequent stages. This ensures coordinated planning, reduces manual intervention, and allows independent module enhancements without affecting overall system performance.

IV. Results and Analysis

A. System Evaluation Overviews

The system was evaluated on its ability to generate feasible residential layouts, optimize interior/exterior spaces, and assess Vastu compliance, focusing on functionality, response time, interoperability, and practical usability rather than isolated accuracy metrics.

Test scenarios included rectangular and irregular plots, space-constrained parcels, and varied user requirements for rooms, furniture, and exterior areas to validate reliability under realistic conditions.

B. Land Utilization and 2D Plan Generation Results

The system produced feasible, optimized 2D floor plans for all tested land configurations, dynamically adjusting room allocations when requirements exceeded land limits, preserving functional integrity.

Layouts maintained logical room arrangements, circulation, and boundary compliance. JSON-based data exchange ensured seamless integration with subsequent modules, demonstrating efficient modular interoperability.

1) Experimental Setup for Land Utilization and 2D Plan Generation: Evaluation used multiple land parcels (150-400 m², rectangular and irregular) with varying user requirements. Optimization employed a genetic algorithm, encoding room positions, dimensions, and spatial relationships. Fitness considered boundary adherence, non-overlapping rooms, minimum dimensions, and space utilization. Each scenario was run multiple times to assess stability and average execution time.

TABLE I
Land Feasibility and Room Placement Results

Case	Area	Shape	Req.	Placed	Feas. (%)	Cov. (%)	Time (s)
Small	150	Rect.	6	5	83	72	2.3
Medium	250	Rect.	8	8	100	65	3.1
Large	400	Rect.	10	10	100	58	4.2
Irregular	200	L	7	6	86	68	2.8
Avg.	250	–	7.75	7.25	**92.3**	**65.8**	**3.1**

2) Land Feasibility and Room Placement Results: The feasibility rate is computed as the ratio of successfully placed rooms to the number of rooms requested by the user. When land constraints prevented full placement, the optimizer intelligently reduced room counts while preserving critical functional spaces.

TABLE II
Space Utilization Comparison Between Random and GA-Based Layouts

Case	Area	Random	GA	Gain	Fitness
1	200	45	68	+51%	87.3
2	300	52	71	+37%	89.1
3	150	38	64	+68%	82.4
4	400	48	69	+44%	91.2
Avg.	262.5	45.8	68.0	**+48.3%**	**87.5**

3) Space Utilization Optimization Performance: The percentage improvement in land utilization is calculated as:

$$\text{Improvement} = \frac{\text{Cov}_{GA} - \text{Cov}_{Rand}}{\text{Cov}_{Rand}} \times 100 \qquad (2)$$

TABLE III
Building Regulation and Constraint Compliance

Constraint	Tests	OK	Viol.	Success (%)	Penalty
No Overlap	20	20	0	100	0.0
Boundary	20	19	1	95	2.1
Setback	20	18	2	90	1.8
Min Size	20	20	0	100	0.0
Overall	80	77	3	**96.3**	**1.0**

4) Constraint Satisfaction Validation: Across all test cases, the genetic algorithm demonstrated stable convergence behavior, achieving high-quality solutions within early generations. On average, valid layouts were produced within 3.1 seconds using a population size of 50 and 100 generations. Solution stability across repeated runs exceeded 89%.

C. Vastu Detection and Compliance Evaluation Results

The module was validated for its ability to detect functional rooms from 2D floor plans and evaluate Vastu compliance based on directional rules defined by Vastu Shastra. 1) Experimental Setup for Vastu Detection and Compli- ance Evaluation: Evaluation used 120 annotated residen- tial floor plans (generated and manual) with an 80:10:10 train-validation-test split and over 500 room instances across 13 categories. Room detection employed a YOLOv8 model with GPU acceleration, while rule-based Vastu evaluation used structured JSON. Results focus on four Vastu-critical rooms; despite moderate detection accuracy from similar layouts, perfor- mance

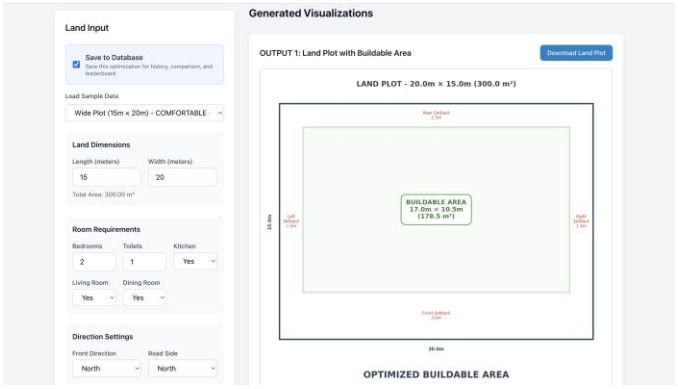


Fig. 6. Sample 2D floor plan generated by the land utilization and GA-based optimization module

was sufficient for reliable directional and compliance assessment. Standard precision and recall metrics were used to evaluate detection performance.

Table IV presents the quantitative performance of the room detection process on the test dataset.

TABLE IV
YOLO-BASED ROOM DETECTION PERFORMANCE

Room Type	Precision (%)	Recall (%)
Bedroom	39.29	28.57
Kitchen	37.04	19.23
Toilet	29.82	16.35
Living Room	49.09	47.37
Overall Average	**38.31**	**27.88**

The YOLO room detection module supports rule-based Vastu evaluation rather than high-precision classification. Despite layout similarity and symbol ambiguity, partial detections provide enough spatial cues for reliable assessment.

TABLE V
VASTU COMPLIANCE EVALUATION AND RULE VIOLATIONS

Compliance Metric	Result	Violation Type	Count
Plans evaluated	20	Kitchen direction	6
Fully compliant	11	Bedroom direction	4
Partially compliant	7	Toilet placement	3
Non-compliant	2	Pooja room location	2
Avg. rules/plan	14		

Results confirm the Vastu module provides consistent, automated compliance, serving as a decision-support tool rather than a standalone vision system.

D. Compliance Score Computation

An overall Vastu compliance score is computed using a weighted aggregation of individual rule evaluations:

$$\text{Score} = \left(\frac{\sum_{i=1}^{N} S_i \times W_i}{\sum_{i=1}^{N} W_i} \right) \times 100 \qquad (3)$$

where S_i represents the compliance impact of rule i (1 for compliant, 0.5 for acceptable, 0 for non-compliant), W_i denotes the importance assigned to rule i, and N is the total number of evaluated rules.

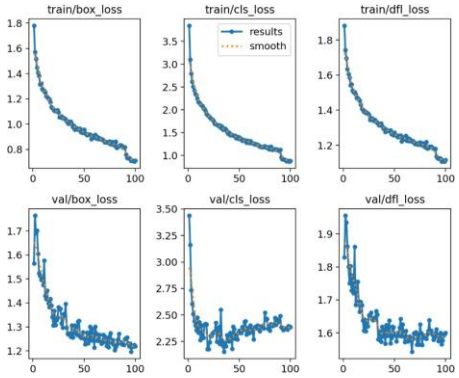


Fig. 7. Training and validation loss curves of the YOLOv8 model for room detection, including box loss, classification loss, and distribution focal loss (DFL).

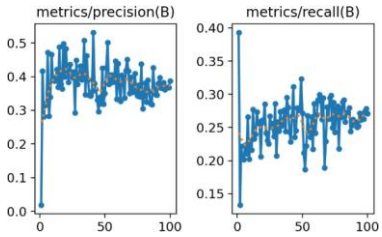


Fig. 8. Precision and recall trends of the YOLOv8 room detection model across training epochs.

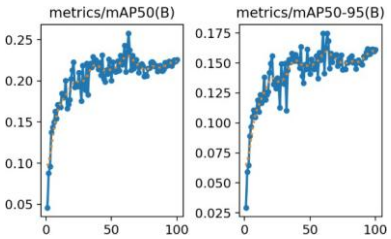


Fig. 9. Mean Average Precision (mAP@0.5 and mAP@0.5:0.95) of the YOLOv8 model for room detection evaluation.

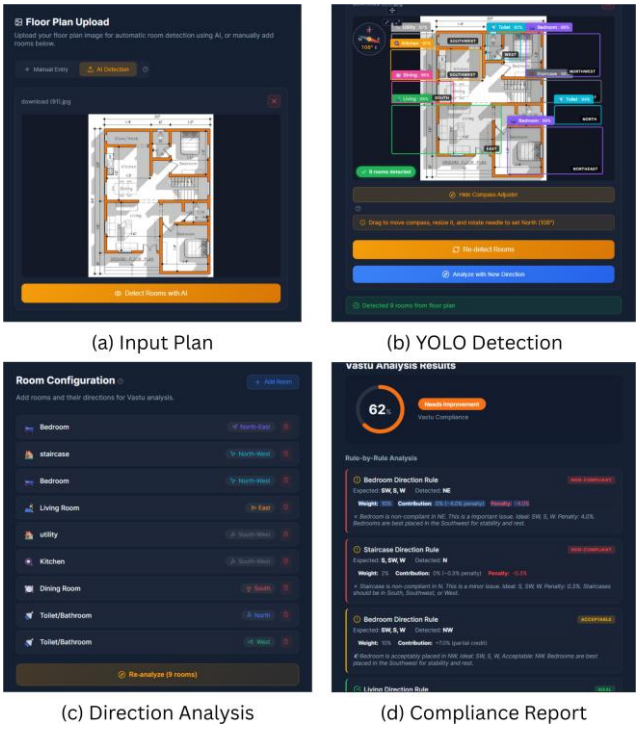


Fig. 10. Sample Vastu detection and compliance evaluation results: (a) input floor plan, (b) detected rooms using YOLO, (c) directional analysis, and (d) generated compliance report

E. Interior Space Optimization Results

The module generated 2D furniture layouts that maximized usable space while ensuring non-overlap, circulation, and accessibility for quick user interpretation.

1) Experimental Setup for Interior Space Optimization: The system was evaluated for valid, collision-free, and aesthetically pleasing layouts across varied room geometries and furniture densities, testing the Genetic Algorithm (GA) and hybrid fitness function.

Test Environment: Implemented in Python with *Shapely* for geometry operations and *PyTorch* for ML-based aesthetic scoring on a standard workstation.

Test Scenarios: The system was tested on distinct room configurations: a $4m \times 5m$ bedroom with bed, wardrobe, desk, chair; and a $4m \times 4m$ living room with sofa, coffee table, TV stand, bookshelf.

hybrid fitness function as defined in Equation (1), where $F_{geo}(L)$ enforces geometric constraints and $S_{ML}(L)$ provides learned aesthetic guidance.

The interior space optimization module successfully generated feasible furniture layouts for bedroom scenarios ranging from 9 m² to 18 m², with average execution times of 3-5 seconds using a genetic algorithm with 50 individuals over 30 generations. The genetic algorithm reliably produced valid layouts for rectangular rooms, demonstrating consistent performance across different room sizes.

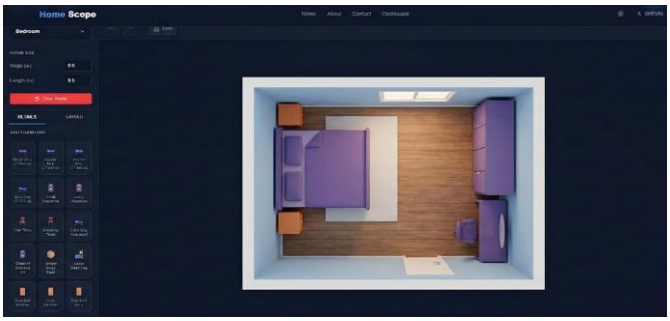


Fig. 11. Optimized furniture layout prototype for a residential interior space

F. Exterior Space Planning Results

The module efficiently allocated outdoor functional areas while ensuring collision-free layouts, spatial feasibility, accessibility, and integration with interior designs. Test scenarios included land parcels of 200-1000 m² with varied footprints and features (parking, gardens, pathways, water elements, pergolas, green roofs), with all layouts respecting boundaries, setbacks, and functional zoning.

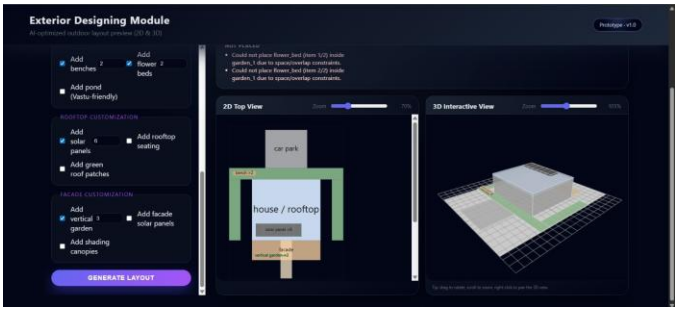


Fig. 12. Optimized exterior space layout illustrating functional zone allocation such as parking, pathways, and green areas.

G. Integrated System Performance

Using centralized JSON-based data exchange, all modules operated cohesively, enabling real-time coordination across land planning, interior and exterior optimization, and Vastu compliance evaluation. The system demonstrated consistent performance with minimal response times, providing users with automated, cost-effective planning tools.

H. Summary of Findings

Experimental evaluation confirmed that the AI-integrated system addresses major limitations of conventional residential planning by combining optimization, analytics, and culturally informed compliance, reducing dependence on external experts and improving planning efficiency.

V. Discussion

The framework supports data-driven decision-making through integrated land utilization, floor plan generation, interior and exterior optimization, and Vastu compliance evaluation. The modular design enables coordinated planning and reduces fragmented manual workflows.

Current limitations include reliance on 2D representations and rule-based cultural evaluation. Future work will focus on 3D visualization, adaptive learning-based optimization, and validation using real-world architectural data.

VI. Conclusion

This study presents a modular AI-based smart home design system combining land utilization, automated 2D floor plan generation, interior and exterior optimization, and Vastu compliance. Structured JSON data flows enable efficient module interaction and scalability. Experimental results show the system produces feasible, optimized residential layouts under varying land constraints, reducing time and expert reliance, offering a practical and cost-effective solution for space- or culture-sensitive planning.

References

[1] L. Wang et al., ”Automated building layout generation using deep learning,” IEEE Int. Conf. Smart Buildings, 2023.

[2] K. Nauata et al., ”House-GAN: Relational generative networks for house layouts,” ACM Trans. Graphics, vol. 39, no. 6, 2020.

[3] J. Redmon et al., ”You Only Look Once: Unified, real-time object detection,” CVPR, pp. 779–788, 2016.

[4] A. Chattopadhyay et al., ”Structured graph VAE for indoor furniture layout,” IEEE Trans. Vis. Comput. Graphics, vol. 28, no. 6, 2022.

[5] H. Tanasra et al., ”Automation in interior space planning using CGANs,” Comput.-Aided Des., vol. 155, 2023.

[6] A. Smith and R. Johnson, ”Constraint-based furniture placement optimization,” J. Build. Perform. Simul., vol. 15, no. 2, 2022.

[7] P. Gupta and V. Sharma, ”Computational Vastu Shastra for residential layouts,” Int. J. Archit. Comput., vol. 18, no. 4, 2022.

[8] L. Wang et al., ”ActFloor-GAN: Adversarial floorplan design,” Autom. Constr., vol. 130, 2021.

[9] J. Kakooee and D. Dillenburger, ”Deep RL for space layout design,” J. AI Archit., vol. 2, no. 1, 2024.

[10] MDPI, ”Automated room usage semantics,” Buildings, vol. 15, 2025.

[11] IJDAR, ”Survey of floor plan image analysis,” Int. J. Doc. Anal. Recognit., 2025.

[12] G. Jocher et al., ”YOLOv8: Ultralytics YOLO,” 2023.

[13] IJDAR, “A comprehensive survey of floor plan image analysis,” International Journal on Document Analysis and Recognition, 2025.

[14] Springer, “Overview of AI in building sustainability assessment,” Springer Briefs in Architecture and Civil Engineering, 2025. [Online]. Available: https://github.com/ultralytics/ultralytics